

Sales Performance Analysis

Prepared for Alfido Tech | Task 2 of 3 — Data Analytics Internship Program

Executive Summary

This report analyzes 9,800 sales line items across 4,922 orders from 793 customers (Jan 2015 – Dec 2018), covering \$2.26M in total sales. Sales grew overall across the period: after a slight dip in 2016 (–4.3%), growth accelerated sharply in 2017 (+30.6%) and continued in 2018 (+20.3%). The business shows strong, predictable seasonality — September, November, and December are consistently the strongest months, driven by back-to-school and holiday buying. **Technology** generates the most revenue (36.6% of sales) despite the fewest orders, while **Office Supplies** drives the most order volume at a much lower average value. Revenue is broadly distributed across the customer base (it takes ~50% of customers to reach 80% of revenue) — a healthier spread than typical retail concentration. Shipping is dominated by the slowest option (Standard Class, 59.8% of orders, ~5 day average delay). Five actionable recommendations are provided in Section 7.

Total Sales	Orders	Customers	Products	2018 YoY Growth
\$2.26M	4,922	793	1,861	+20.3%

Key Findings

- Sales grew overall across the period — a 2016 dip (–4.3%) was followed by strong growth in 2017 (+30.6%) and 2018 (+20.3%).
- Strong, predictable seasonality — September, November, and December are peak months, each 2–3x a typical mid-year month.
- Technology drives 36.6% of revenue from the fewest orders (highest revenue-per-order); Office Supplies drives the most order volume (60% of line items) at the lowest average value.
- Revenue is broadly spread across customers — it takes ~50% of customers to reach 80% of revenue, healthier than typical 80/20 concentration.
- Standard Class shipping dominates (59.8% of orders) despite being the slowest option (~5 day average delay vs. <1 day for Same Day).

Top 5 Recommendations (see Section 7 for full detail)

1. Plan inventory and staffing around the Sep/Nov/Dec seasonal peak.
2. Promote faster shipping tiers more assertively at checkout.
3. Double down on Technology as a margin/value driver via bundling and upsell.
4. Use Office Supplies as a traffic driver, then cross-sell higher-value categories.
5. Investigate and target underperforming states for incremental revenue.

1. Introduction & Data Overview

Dataset: Superstore Sales Dataset (Kaggle) — *superstore_final_dataset.csv*. The file contains line-item-level sales records: one row per product sold within an order, with customer, geography, product, and shipping details.

Column	Description
Order_ID / Order_Date / Ship_Date	Order identifier and dates (DD/MM/YYYY)
Ship_Mode	Standard Class, Second Class, First Class, or Same Day
Customer_ID / Customer_Name / Segment	Customer identity and segment (Consumer, Corporate, Home Office)
Country / City / State / Region / Postal_Code	Shipping geography
Product_ID / Category / Sub_Category / Product_Name	Product identity and classification
Sales	Line-item revenue (used as the primary metric throughout)

Scope: 9,800 line items across 4,922 orders, 793 customers, and 1,861 products, spanning 2015-01-03 to 2018-12-30. Note: this dataset variant includes only the Sales field — Profit, Quantity, and Discount (present in some Superstore variants) are not available here, so this analysis focuses on revenue performance rather than profitability.

2. Data Cleaning & Methodology

- **Date format:** Order_Date and Ship_Date are in DD/MM/YYYY format (confirmed by checking for day values above 12 in the first slot) — parsed with day-first interpretation.
- **Duplicate rows:** checked, none found.
- **Missing values:** Postal_Code missing for 11 rows, all in Burlington, Vermont (a known real-world ZIP quirk) — left as-is since Postal_Code isn't used in this analysis.
- **Invalid values:** checked for non-positive Sales; none found.
- **Date consistency:** confirmed Ship_Date is never before Order_Date for any row.
- **Feature engineering:** extracted Order_Year, Order_Month, Order_Quarter, Order_Weekday, and Shipping_Delay_Days (Ship_Date – Order_Date) for trend and shipping analysis.

3. Sales Trend Analysis

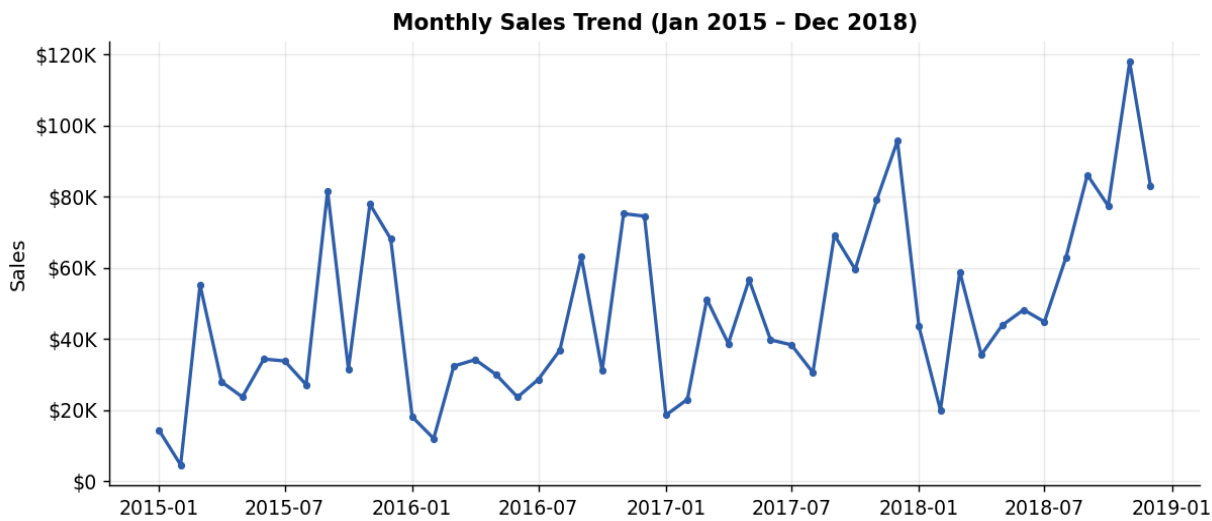


Figure 1. Monthly sales trend, Jan 2015 – Dec 2018. Clear upward trajectory with a pronounced annual peak around November.

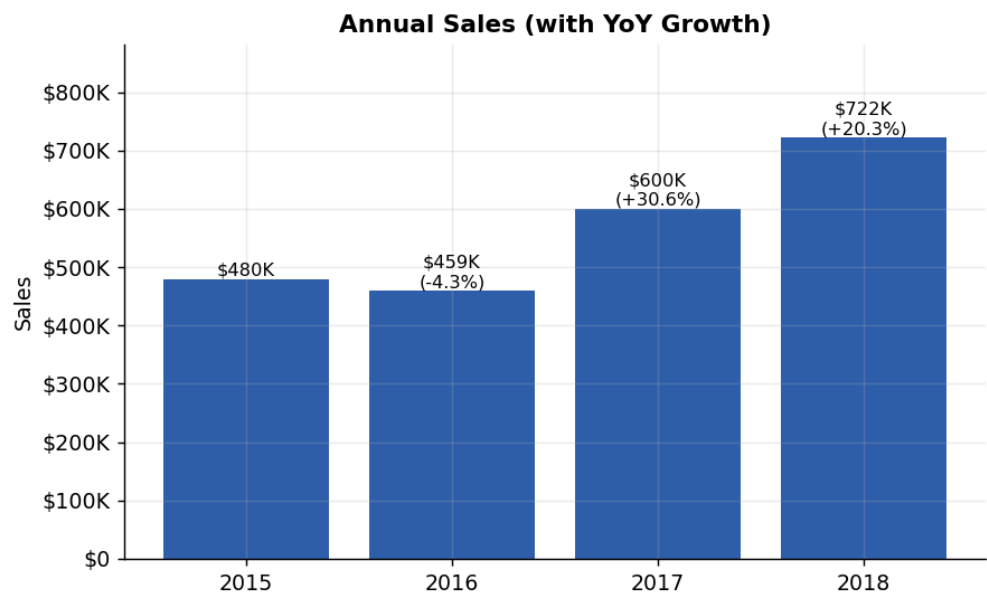


Figure 2. Annual sales with year-over-year growth. 2016 saw a slight contraction, followed by strong recovery and growth.

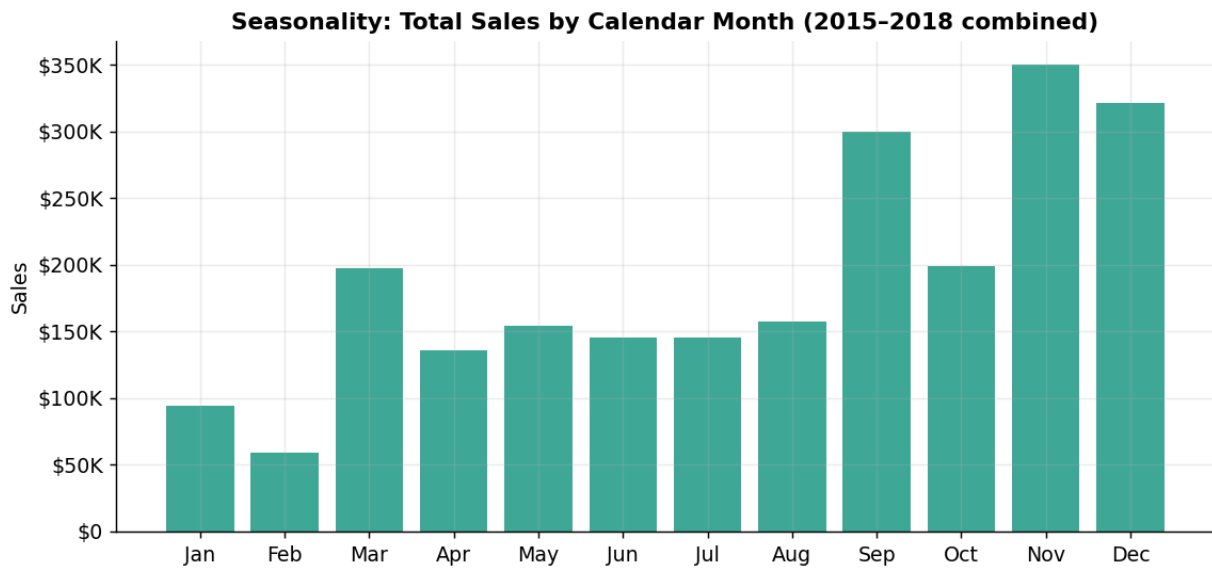


Figure 3. Seasonality — total sales by calendar month, summed across all four years. September, November, and December stand out clearly.

4. Category, Sub-Category & Product Performance

Category	Total Sales	Orders	Avg. Order Value	% of Revenue
Technology	\$827,456	1,813	\$456.40	36.6%
Furniture	\$728,659	2,078	\$350.65	32.2%
Office Supplies	\$705,422	5,909	\$119.38	31.2%

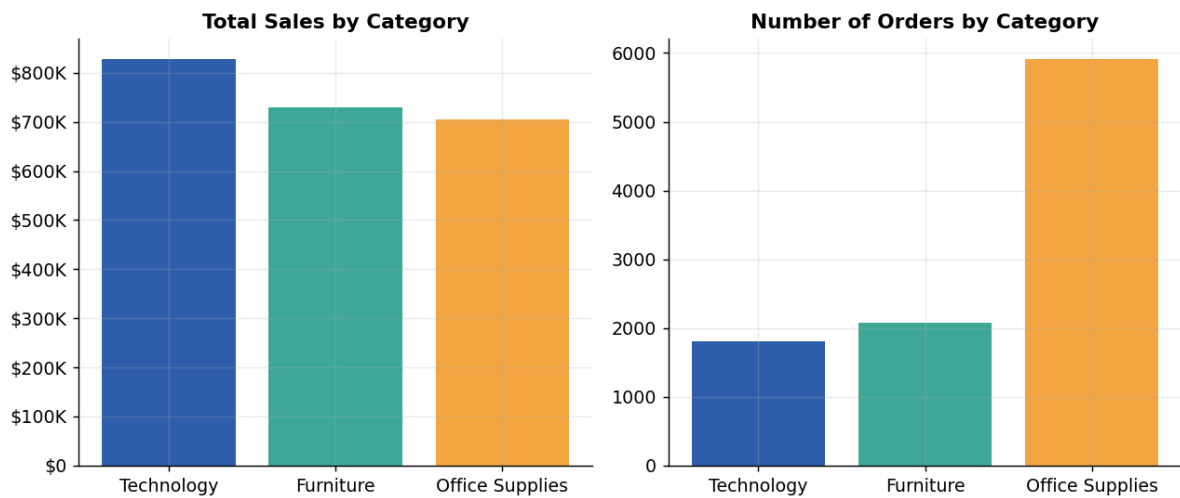


Figure 4. Sales and order count by category. Technology earns the most per order; Office Supplies wins on volume.

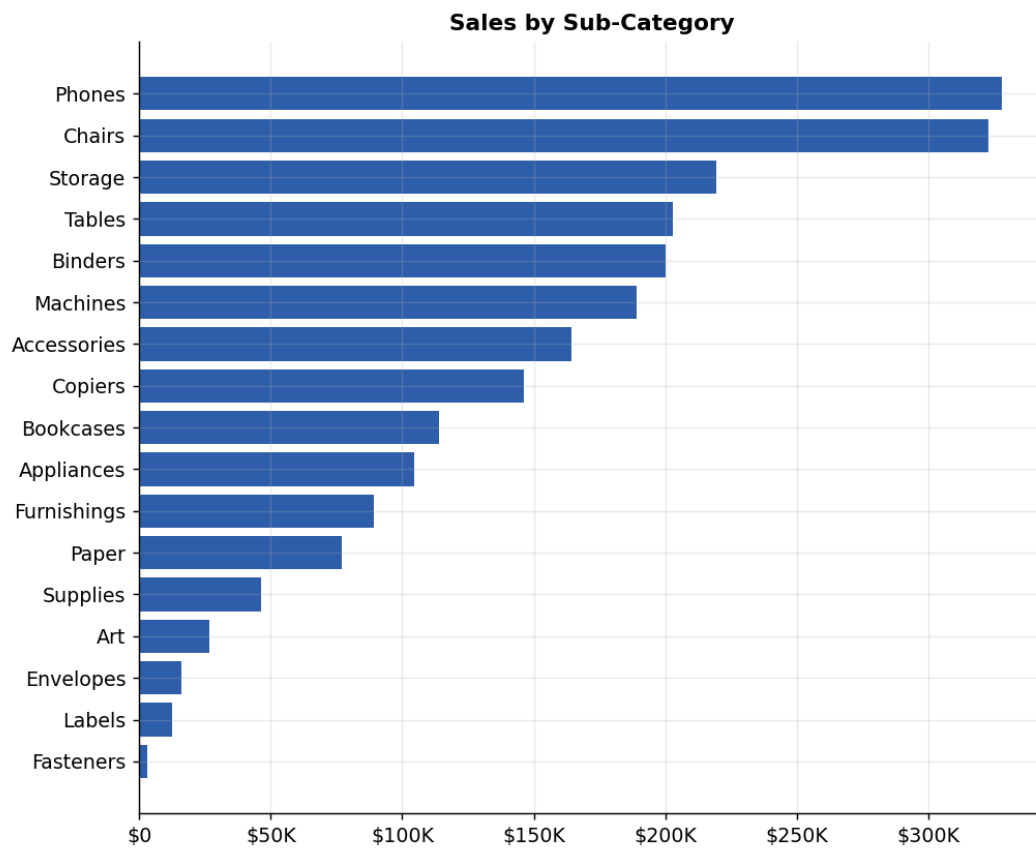


Figure 5. Sales by sub-category. Phones, Chairs, and Storage lead; Fasteners, Labels, and Envelopes trail far behind.

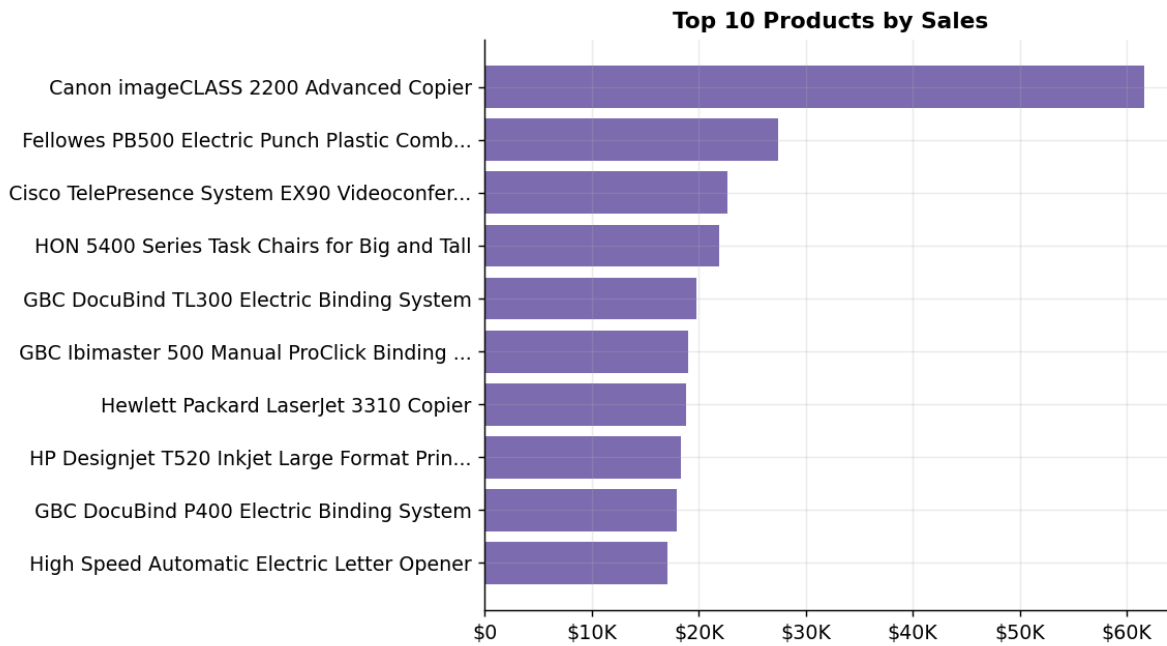


Figure 6. Top 10 products by total sales, led by the Canon imageCLASS 2200 Advanced Copier at over \$61K.

5. Regional & State Performance

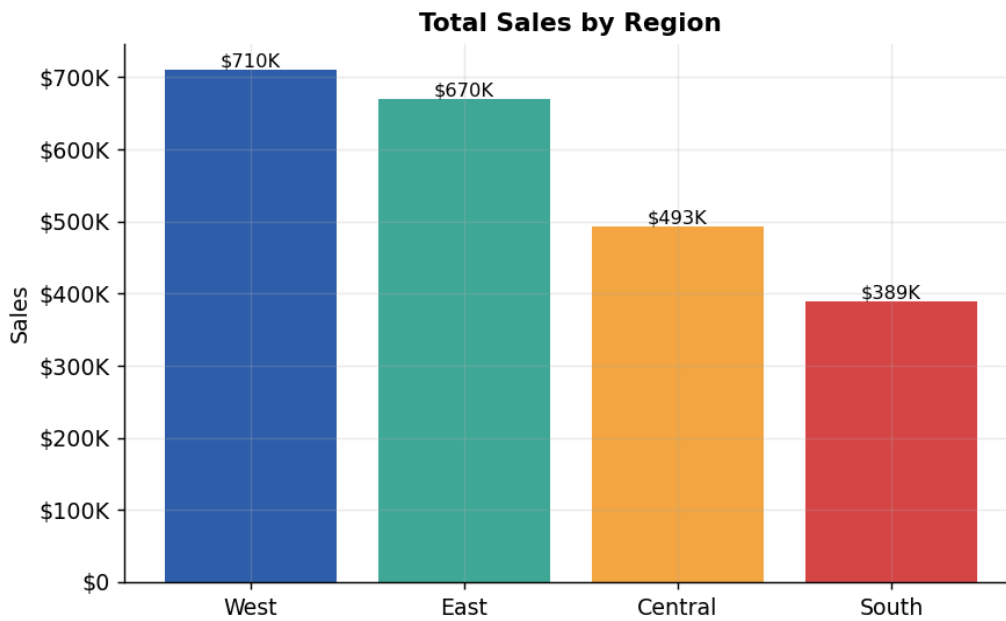


Figure 7. Total sales by region. West (31.4%) and East (29.6%) lead; South (17.2%) trails.

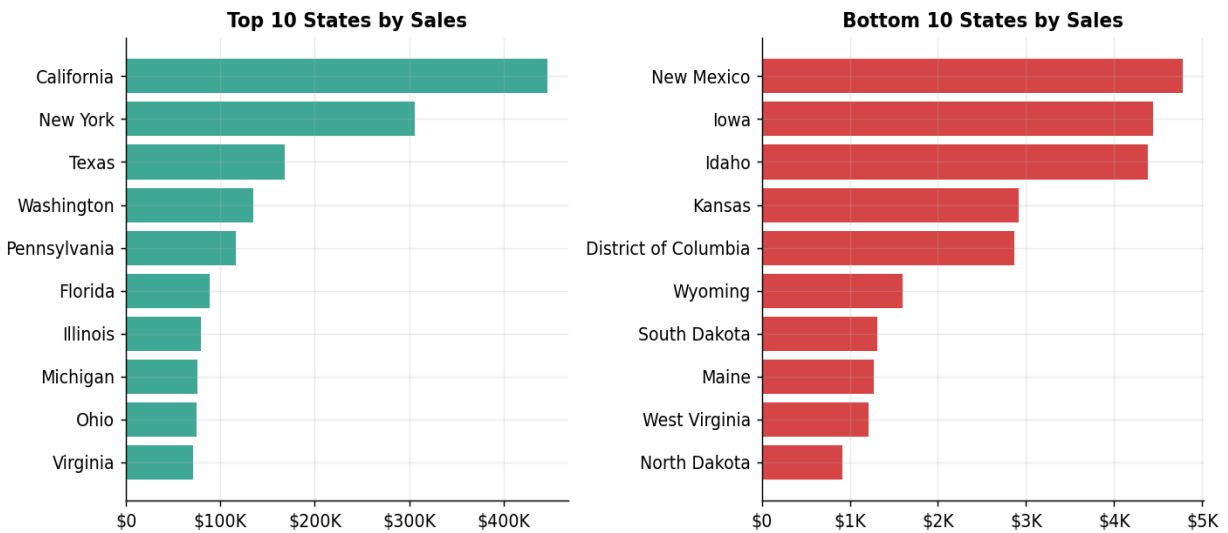


Figure 8. Top and bottom 10 states by sales. California and New York dominate; several small/rural states contribute under \$2K each over the full 4-year period.

6. Customer Segment & Concentration Analysis

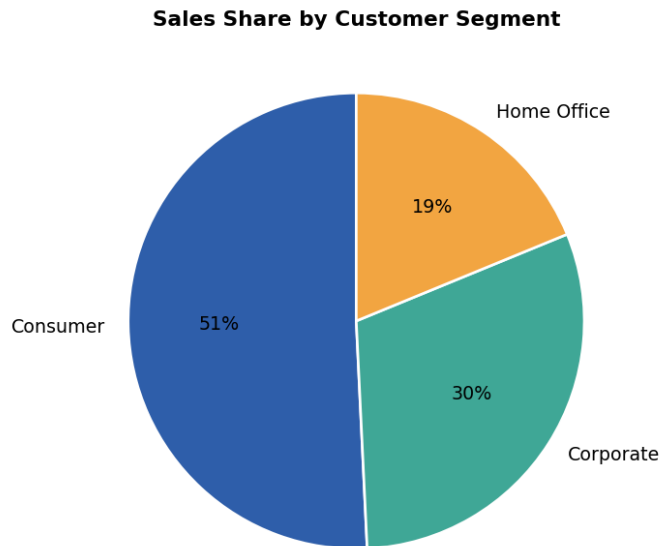


Figure 9. Sales share by customer segment. Consumer leads at 50.8%, followed by Corporate (30.4%) and Home Office (18.8%).

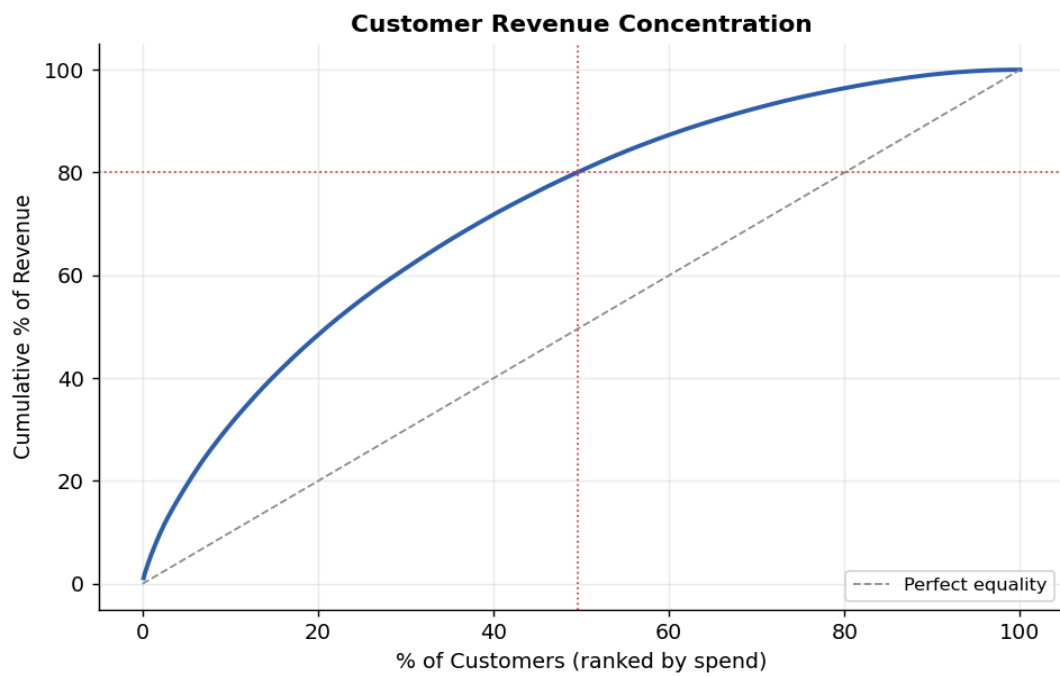


Figure 10. Customer revenue concentration curve. It takes ~50% of customers to reach 80% of revenue — notably less concentrated than a classic 80/20 pattern, indicating a broad, healthy customer base rather than dependence on a small handful of accounts.

7. Shipping Mode Analysis

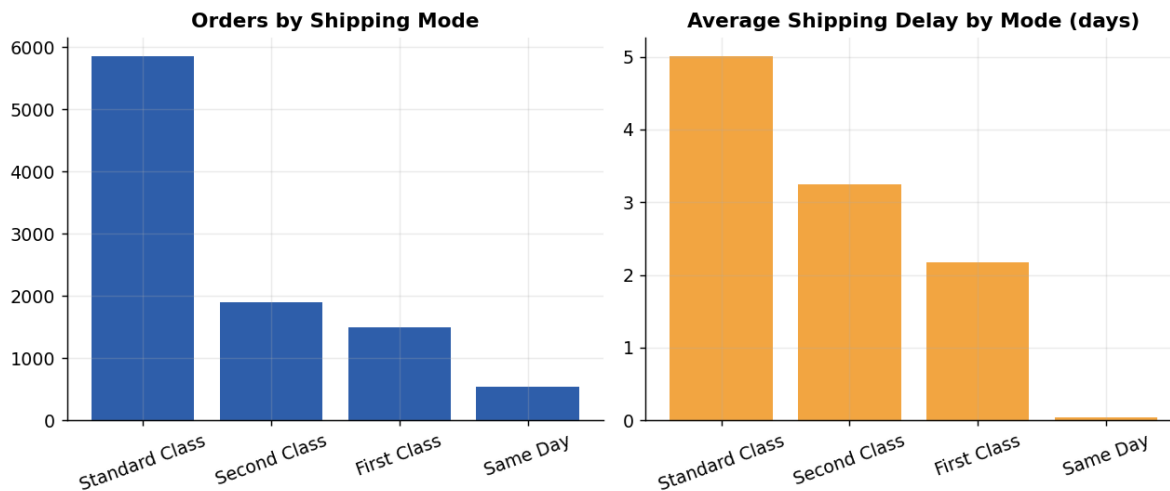


Figure 11. Orders by shipping mode and average delay. Standard Class accounts for 59.8% of orders but has the longest average delay (~5 days); Same Day ships in under a day but is used for only 5.5% of orders.

This gap — most customers defaulting to the slowest shipping option — could reflect price sensitivity (if faster options cost more), a lack of visibility into faster options at checkout, or simply that most orders aren't time-sensitive. Any of these would be worth validating with Alfido Tech's checkout data before investing in shipping changes.

8. Actionable Recommendations for Alfido Tech

1. Plan inventory and staffing around the seasonal peak

September, November, and December together account for a disproportionate share of annual revenue. Build stock, staffing, and marketing pushes ahead of these three months specifically, rather than spreading resources evenly across the year.

2. Promote faster shipping tiers more assertively

With 59.8% of orders on the slowest Standard Class option, testing discounted or bundled upgrades to First Class or Second Class at checkout could improve customer satisfaction and potentially unlock willingness-to-pay for speed, without materially increasing fulfillment risk.

3. Double down on Technology as a margin/value driver

Technology's high revenue-per-order (\$456 average vs. \$119 for Office Supplies) makes it a strong candidate for bundling, upselling, and premium placement in marketing, even though it generates fewer orders than other categories.

4. Use Office Supplies as a traffic driver, then cross-sell

Office Supplies' high order count but low average value makes it a good low-friction entry point. Pairing Office Supplies purchases with Technology or Furniture recommendations at checkout could lift overall basket size without needing to acquire new customers.

5. Investigate and target underperforming states

States like North Dakota, West Virginia, Maine, and South Dakota each contributed under \$2,000 across the full four-year period. Targeted regional marketing, localized promotions, or a review of distribution/shipping costs to these areas could unlock incremental revenue in the long tail.

Prepared as part of the Alfido Tech Data Analytics Internship Program — Task 2: Sales Performance Analysis. Full code, methodology, and additional exploratory visualizations are available in the accompanying Jupyter notebook (Sales_Performance_Analysis.ipynb).